SAGE: IRL

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***Abstract*—Public Large Language Models (LLMs) are being increasingly explored for aiding students to learn difficult concepts in STEM education. However, such systems are usually told what a student knows rather than judging it, and treat a topic as learned because a course was completed or an instructor marked it off. Those that do judge reduce every quiz, exercise, and program a student has completed to a single score. A student who answers factual questions well but cannot yet write working code is then indistinguishable from one who can do both, a failure we term the Evidence Diversity Problem. In this paper, we present a learner model for "SAGE" (Student-focused Adaptive Guidance Engine), an intelligent tutoring framework that enforces instructor-regulated learning. The model tracks each topic as three separate estimates covering recall, step-by-step execution, and applied work on a task of the student's own, and marks a topic as learned only when the weakest of the three is satisfied. We prove that certain patterns of evidence cannot mark a topic as learned at any setting of the model's parameters. We evaluate SAGE on 2000 simulated learners per condition, a custom educational benchmark, a held-out multi-turn adversarial audit, and task-based sessions with human participants. Single-score tracking marks 1998 of 2000 learners who are strong on recall and weak in applied work as having learned the topic, and demanding more evidence without separating its kinds changes almost nothing at 1994, while our rule marks none. An existing multi-dimensional model also marks none of these learners, so this case does not separate the two rules. The difference appears on a learner who is middling at every kind of work, where that model lets two adequate scores make up for a poor third. Our rule requires each kind to stand on its own and marks 21% fewer such learners as having learned the topic. Curriculum compliance rises from 80% to 100% over the conference system, and the session-level enforcement layer deflects 99.4% of adversarial turns across 40 attack sessions while containing every restricted-asset request.**

***Index Terms*—Human-AI Teaming, Intelligent Tutoring, Knowledge Graphs, Cognitive Scaffolding, Access Control**

I. Introduction

The rapid proliferation of LLMs has transformed the landscape of Intelligent Tutoring Systems (ITS). Unlike earlier rule-based or scripted tutors, public LLMs such as ChatGPT can explain concepts, debug code, and synthesize examples in real time, creating the promise of scalable, personalized instruction for millions of learners [1]. At the same time, their widespread use in academic settings has raised unprecedented concerns about academic integrity, over-reliance on automation, and the hindrance to genuine learning habits/processes [1], [2].

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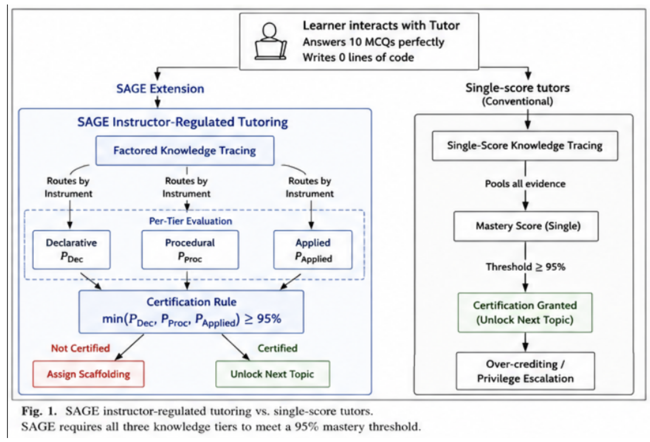

Fig. 1: SAGE instructor-regulated tutoring against single-score tutors. A learner who answers ten multiple-choice items correctly and writes no code is certified by a single-score rule and refused by the factored rule, which requires all three tiers to meet the same bar.

LLMs are optimized to produce fluent, direct and complete answers. Effective instruction seeks the opposite: Instructor-Regulated Learning (IRL), in which instructors plan, monitor and assess what a learner is ready for [3]. When an LLM returns an end-to-end solution to a task such as "How do I reverse an array?", the learner consumes an answer rather than engaging with the reasoning that produces it, and may lack the background the answer presumes.

Current AI-tutoring systems connect LLMs to course material through Retrieval-Augmented Generation (RAG) [4], but treat the integration as a search task. As illustrated in Figure **??**, an open RAG pipeline retrieves the most semantically similar content whether it is a practice problem, an instructor solution key, or an advanced topic the learner has not yet reached. This disregards curriculum structure and learner readiness, and it introduces a security vulnerability: sensitive instructional materials such as answer keys or exam papers can be exposed through ordinary dialogue [5]. Thus, what appears as a retrieval design choice simultaneously becomes a pedagogical failure and a instruction policy breach.

Prior research does not treat learning goals and security measures as connected issues. Educational practice emphasizes *soft alignment* through prompts that ask the model to "act as a tutor" [1] [2], while security mechanisms emphasize *hard restrictions* that block access to protected data [5]. This separation leads to undesired consequences: prompts can be ignored or manipulated by students seeking answers, while rigid blocking mechanisms cannot distinguish between legitimate scaffolding and cheating. We argue that in AI-assisted pedagogical systems, there is a gap in the state-of-the-art in terms of understanding how to configure access control rules by having curriculum structure being a critical part of the system's security posture.

Our prior system, SAGE [6], addressed the instructor's side of this problem. Rather than relying on an LLM to refuse requests through prompt engineering, SAGE enforces instructional policy on a standard RAG pipeline through a three-layer Safety Gatekeeper placed between the learner and the knowledge base: i) Role-Based Access Control (RBAC): blocks access to unauthorized files, ii) Attribute-Based Access Control (ABAC): aligns content access with curriculum stage and learner progress, and iii) Cognitive Access Control (CAC): infers learner intent and favors hints and scaffolding over full solutions. Composed over a knowledge graph of learner state and course content, this gatekeeper reached 90% security and 80% curriculum compliance while remaining competitive in helpfulness against prompt-based tutors [6].

SAGE thus provides dependable access control over a set of topics the learner is treated as having mastered, but supplies no principled mechanism for deciding what belongs in that set. Membership is conferred by course completion or instructor status rather than earned through evidence of learning, so the attribute on which every access decision turns, i.e., the learner's mastered set, is asserted rather than verified.

Deciding what a learner has genuinely mastered is harder than it first appears, because mastery of a programming concept is many-sided. Recalling what a pointer is, tracing by hand how one behaves in a loop, and writing correct code that uses it are distinct competencies, and a learner may hold one without the others. A tutor that certifies mastery from a single kind of evidence, e.g., multiple-choice recall alone, over-credits learners who have never demonstrated the rest. We term this the *Evidence Diversity Problem*, i.e., the systematic over-crediting that arises when evidence of different kinds is collapsed into one mastery estimate. Under instructor-regulated learning it corrupts the prerequisite relations the course enforces.

In this article, we extend SAGE with a learner model that turns a stream of mixed interactions into a record of concepts certified by evidence, and feeds that record directly into the existing gate. This article makes the following contributions:

- We name the Evidence Diversity Problem and show that it carries a consequence specific to instructor-regulated tutoring. Over-crediting from one kind of evidence is a familiar measurement failure [7]. When a mastery record governs curriculum access, however, an inflated record becomes an escalation of privilege, since the learner obtains material the course withheld on the strength of competence they were never shown to possess.
- We introduce a learner model that factors a single concept by mode of demonstration rather than by component skill. Recall, procedural execution, and applied problem-solving on the same topic are tracked as three separate estimates, each fed by its own instrument under a fixed routing rule. Prior work factors by skill; we factor one skill by the kind of evidence that demonstrates it. Certification reads the weakest of the three rather than their product, so strength in one kind of evidence never offsets a deficit in another, at any threshold.
- We prove an Evidence Diversity Guarantee. Certain patterns of evidence cannot certify a concept at all, however much evidence they contain and whatever values the model's parameters take: a concept requires a minimum number of observations at every level, and a level with no correct observation can never be satisfied. The guarantee follows from the architecture rather than from any configuration.
- We integrate the model into SAGE so that its prerequisite gate acts on earned rather than asserted mastery, and evaluate the composed system: a response policy that fades scaffolding as competence grows, verified by showing that a blind judge can recover the assigned level from response text alone, and a session-level enforcement layer under which no session is refused without conversation-level adjudication, whose contribution is isolated by ablation on a held-out adversarial suite.

Our evaluation isolates the learner model in a simulation harness, replays a fixed educational benchmark through the deployed system, audits the session-level layer against multi-turn attacks, and runs task-based sessions with human participants under IRB approval. Single-score tracking certifies 1998 of 2000 learners who are strong on recall and weak in applied work, where our rule certifies none. Against an existing multi-dimensional model our rule certifies 21% fewer under-prepared learners where compensation between kinds of evidence is available. Curriculum compliance rises from 80% to 100% over the conference system, and the session-level layer deflects 99.4% of adversarial turns.

The remainder of this article is organized as follows. Section II reviews related work. Section III recaps the inherited architecture and formalizes the learner model, the certification rule, and the Evidence Diversity Guarantee. Section IV presents the evaluation, including simulation, benchmark replay, adversarial audit, and participant sessions. Section VI concludes.

II. Related Work

A. Intelligent Tutoring Systems

Regulated learning is a critical component for academic success, traditionally viewed through Zimmerman's cyclical model [3]. It consists of learnable processes, such as setting goals and self-monitoring, that are heavily influenced by instructional scaffolding and feedback [8], [9]. Early ITS utilized rule-based pedagogical policies to explicitly prompt learning behaviors such as help-seeking and metacognitive monitoring. While these scaffolds effectively improve learning outcomes and regulatory skills [10], [11], their reliance on manual authoring and rigid domain modeling limits scalability. Conversely, LLMs offer scalable, personalized instruction through natural-language interaction [1]. However, the shift toward generative AI-based tutors necessitates new mechanisms to ensure these systems actively support, rather than bypass, a learner's internal regulatory processes.

Further, recent studies caution that general-purpose LLMs are not inherently aligned with pedagogical objectives as they prioritize providing complete answers over instructional value [1]. This tendency can allow learners to bypass the productive struggle and metacognitive engagement essential for instructor- or self-regulated learning. Empirical evidence also confirms that the pedagogical impact of AI chatbots depends heavily on interaction design. Towards this, a meta-analysis indicates that learning outcomes improve when interactions provide guided support; conversely, unguided answer-giving can negatively affect higher-order problem-solving [2]. Moreover, systematic reviews suggest that while LLMs enhance accessibility, they risk fostering a reliance on the model that diminishes a learner's internal regulatory processes when deployed without explicit instructional constraints [12]. We address such concerns in our work through investigations on how interaction structures can be designed to scaffold planning, monitoring, and reflection behaviors, therefore supporting learners' ability to learn when engaging with general-purpose LLM-based tutors.

B. Retrieval-Augmented Generation in Educational Tutoring

To mitigate hallucinations and ground responses in course materials, many tutoring systems utilize RAG [4].

Retrieval grounded in a curated domain corpus measurably improves response relevance for novice users over general-purpose LLMs [13]. Relevance, however, is not pedagogical appropriateness: semantic similarity is agnostic to what a learner is ready for. It fails to account for prerequisite structures or curriculum sequencing; consequently, the most "relevant" retrieved document may be an answer key or advanced material that bypasses the necessary learner's struggle. Beyond pedagogical concerns, the retrieval layer in RAG presents a significant security risk as well. Recent research [5] indicates that retrieved context is a primary leakage channel where prompt injections can manipulate LLMs into revealing sensitive or restricted information once it enters the context window. Since generation-layer safeguards are often probabilistic and bypassable, these risks motivate the enforcement of constraints at the retrieval boundary rather than relying solely on language-level refusals.

Moreover, current tutoring approaches typically treat learning goals and security mechanisms as separate problems. While educational research investigates the potential of LLMs to enhance regulated learning and metacognitive support [1], [2], the security community emphasizes the risks associated with prompt injection and data poisoning in these integrated systems [5]. The disconnect between these domains creates a significant limitation in existing platforms, as the absence of integrated access controls prevents the enforcement of structured, step-by-step guidance when students access RAG implementations for course content. Existing solutions for content moderation often rely on alignment training [14] or explicit regex-based content filters at the generation layer. However, these safeguards do not change underlying LLMs' access to information; therefore, as long as sensitive content is present in the retrieval context, there remains a non-zero probability that it will be exfiltrated under adversarial prompting [5].

In addition to the advances in RAG, recent work on GraphRAG that uses knowledge graphs has demonstrated improved controllability by encoding concept dependencies and curriculum structure that flat vector similarity ignores [15]. There has been further evidence that shows using knowledge graphs can enhance interpretability and alignment in sequenced learning domains [16]. However, existing GraphRAG and secure RAG frameworks typically treat retrieval as a neutral preprocessing step, rather than an explicit component of the instructional design. From a security perspective, this limitation has motivated a shift toward restricting LLM access to data, rather than relying solely on behavioral alignment, to mitigate risks such as prompt injection [5]. The above works motivate our Safety Gatekeeper-based approach in which retrieval itself is governed by pedagogical and organizational policy, enabling access to be conditioned on instructional intent and curriculum constraints.

Enforcement at the retrieval boundary is conventionally applied one message at a time. A per-message decision, however, cannot detect an attack that is assembled across a conversation. The crescendo threat model [17] demonstrates this directly: an adversary decomposes a prohibited goal into individually innocuous turns, and the harmful payload is assembled only by a final instruction that carries no indication of intent on its own. Work on agent memory reaches the same conclusion in a different setting, validating each stored record against the surrounding interaction history because harm that emerges only in context is invisible to record-by-record auditing [18]. Towards this, session-level risk aggregation has been proposed as a training-free defense that escalates suspicious conversations to a conversation-level adjudicator [19]. We adopt this posture in Section III-C, where the unit of enforcement is the session rather than the query.

C. Learner Modeling: Knowledge Tracing and Cognitive Diagnosis

Estimating what a student knows from a sequence of responses is the province of *knowledge tracing*. Bayesian Knowledge Tracing (BKT) [20] models each skill as a two-state hidden Markov process and updates a mastery probability after every response, giving a transparent, online estimate that has anchored mastery-based progression in intelligent tutors for three decades. Deep Knowledge Tracing (DKT) [21] and its successors replace the per-skill Markov assumption with a recurrent network, improving predictive accuracy by sharing statistical strength across skills at the cost of interpretability and a substantial data requirement. Both families share a property that matters here: they summarize competence on a skill as a *single* number, collapsing recall, procedural fluency, and applied problem-solving into one. A learner who answers multiple-choice questions well but cannot yet write working code is indistinguishable, to such a model, from one who can do both, which is the over-crediting the Evidence Diversity Problem describes.

Cognitive diagnosis models (CDMs) take the complementary step of separating competence into several attributes. The DINA model [22], [23], in particular, applies a *conjunctive* rule: a task is answered correctly only if the learner holds *all* of its required attributes, so mastery of a compound skill demands every component at once. CDMs, however, are built for after-the-fact psychometric diagnosis. Their attributes are estimated in batch from a fixed response matrix against a hand-authored Q-matrix, they do not update as a student works, and they do not route evidence by *type*, treating all items that load on an attribute as interchangeable. They tell an analyst which attributes a cohort lacks; they do not hand a tutor a running mastery signal on which to gate the next interaction.

Conjunctive criteria have also been applied within knowledge tracing itself. Conjunctive knowledge tracing [24] requires every skill attached to an item to be mastered before the item is retired, and multi-skill and dynamic-Bayesian variants of BKT jointly model several knowledge components rather than one per model [25]. Closest to our motivation, Huang et al. [26] observe that tracing decomposed component skills alone risks premature assertion of mastery, and grant mastery only when a learner demonstrates a skill in combination with others, evaluated on a Java programming tutor. We compare against [24] directly in Section IV-E. Two things distinguish the present model. The first is the axis along which evidence is factored: prior work factors by knowledge component, i.e., several distinct skills, whereas we factor a single concept by mode of demonstration. The second is the form of the conjunction. A criterion that multiplies per-component estimates still lets two strong components offset a weak third, and raising the product threshold reduces how often this happens without forbidding it. The per-tier minimum of Eq. (18) forbids it at every threshold, which is what an access-control decision requires, since a gate that opens on offset evidence admits a learner on the strength of competence they were never shown to possess.

TABLE I: Capabilities of representative learner-modeling and tutoring approaches. ✓: supported; △: supported only by some variants; ×: not supported. CKT denotes conjunctive knowledge tracing [24].

Capability	BKT	DKT	CDM	CKT	SAGE	Ours
Online updating	✓	✓	△	✓	×	✓
Multi-dimensional mastery	×	×	✓	✓	×	✓
Evidence-type routing	×	×	×	×	×	✓
Conjunctive mastery	×	×	✓	✓	×	✓
Non-compensatory at all thresholds	×	×	△	×	×	✓
Curriculum enforcement	×	×	×	×	✓	✓
Mastery-conditioned adaptation	×	×	×	×	×	✓

D. Mastery-Based Adaptation and Expertise Reversal

Once a mastery estimate exists, a tutor must decide how to act on it. The *expertise-reversal effect* [27] establishes that support which helps novices, e.g., worked examples and heavy scaffolding, becomes redundant or counterproductive as competence grows, motivating adaptation that fades support in step with mastery and keeps instruction within the learner's zone of proximal development [28]. Our mastery-conditioned response policy (Section III) applies this principle to an LLM tutor, mapping the certified level to scaffolding depth, response format, and challenge type. The policy is only as sound as the signal it reads, which is what the learner model of Section III supplies.

E. Summary of the Gap

Table I places these strands side by side. Each capability the Evidence Diversity Problem demands is present in prior work, but never all at once: knowledge tracing updates online yet collapses every kind of evidence into one score, cognitive diagnosis separates dimensions yet diagnoses in batch and ignores evidence type, and instructor-regulated tutors such as SAGE act on a mastered set without determining what belongs in it. The learner model of this article supplies the combination, so that the resulting record can be trusted by an instructor-regulated gate.

III. SAGE Methodology

A fundamental challenge in the deployment of general-purpose LLMs for education is the *alignment gap* between a Large Language Model's stochastic helpfulness objective and the rigid constraints of an academic curriculum. SAGE addresses this by transforming the LLM tutor into a *stateful component* of an ITS, governed by a deterministic Safety Gatekeeper as shown in **Figure ??** that controls what information is available to the LLM's context. This design enables the SAGE to respond naturally to user queries while i) preserving the learner's regulated learning cycle, and ii) strictly enforcing institutional policies regarding assessment content and curriculum pacing. We model the learner session state as:

$$S = \{K, R, C\} \tag{1}$$

where K denotes the set of mastered knowledge nodes (topics), R is the user role (e.g., student or instructor), and C denotes the current cognitive intent (e.g., scaffolding request, solution seeking, or administrative command). Each incoming query q updates S and determines whether access to a candidate document d is permitted. In ITS terms as a human–machine system, the LLM operates as an *actuator* within a closed control loop, where learner actions shape the session state S, thereby constraining what the LLM can retrieve and generate.

A. Dual-Store Knowledge Architecture

SAGE employs a hybrid knowledge architecture that integrates unstructured semantic retrieval with structured curriculum modeling. Unlike flat vector retrieval, a GraphRAG design is used, consisting of two coupled stores: i) A **Vector Database** V (ChromaDB), which indexes chunked instructional content as dense embeddings for semantic similarity retrieval and ii) A **Knowledge Graph** $G = (T, E)$ (Neo4j), where T is the set of topic nodes and E encodes prerequisite relationships in a Directed Acyclic Graph (DAG). Each topic $t \in T$ stores difficulty, instructor-defined learning status, and mastery metadata.

Given a query q, initial candidate chunks are retrieved via:

$$\mathcal{D}_{cand} = \text{topK}(V, \text{embed}(q)). \tag{2}$$

Named entities and topic mentions are mapped to nodes in G. For a topic t, prerequisites are:

$$\text{Prereq}(t) = \{t' \in T \mid (t', t) \in E\}. \tag{3}$$

Given learner knowledge state K, readiness is defined as:

$$\text{Ready}(t, K) = \begin{cases} 1, & \text{if Prereq}(t) \subseteq K, \\ 0, & \text{otherwise.} \end{cases} \tag{4}$$

Thus retrieval becomes *state-aware*: if q requests "arrays" but the learner lacks proficiency in "loops," then $\text{Ready}(\text{Arrays}, K) = 0$ and the system redirects toward prerequisite scaffolding rather than providing a full solution.
Human-in-the-Loop Control: Instructors can modify topic attributes and prerequisite relations within the Knowledge Graph serving as a "human-in-the-loop control", and these modifications are immediately reflected in the Safety Gatekeeper decisions during subsequent queries as shown in **Figure ??**. An instructor-facing interface is used to allow authorized users (i.e., instructors) to lock or unlock topics, adjust prerequisite relationships, and flag sensitive materials within the Knowledge Graph. These updates propagate immediately to the Safety Gatekeeper, enabling real-time policy enforcement without redeploying LLMs or retraining classifiers. This design supports dynamic curriculum management and allows instructors to intervene when pedagogical goals or assessment policies change.

Thus, SAGE operates over an instructor-controlled course corpus containing lecture notes, practice materials, and restricted assessment artifacts. The corpus is chunked into passages for vector indexing, and each chunk d is annotated with: (i) a type label $type(d)$ such as lecture, practice, solution, exam key for RBAC; (ii) an associated curriculum topic t when applicable for ABAC; and (iii) a response mode $mode(d)$ such as conceptual explanation, hint, partial solution, and instructor-only for CAC. These annotations are produced via an instructor-facing interface and validated by spot checks.

Cross-Turn Topic Resolution: State-aware retrieval as formulated in Eq. (4) assumes that each query names the concept it concerns. Real tutoring dialogue does not. Following a definition of arrays, a learner asks "how does it work?", "how do I declare it?", or "what about a nested one?", none of which names a topic. Left unresolved, such a turn either retrieves the wrong concept or reaches the gatekeeper as a context-free fragment and is misread as off-topic. We therefore resolve reference deterministically rather than by inference. The session tracks an anchor t_{anchor}, the most recent concept the learner has named, and a follow-up that names no topic is rewritten against this anchor before any access decision is taken. The rewrite is guarded against an explicit topic switch and against messages matching a small set of frustration patterns, and it runs before the gatekeeper, so that $\text{Ready}(t, K)$ is evaluated on the intended concept. Eq. (4) is not realizable on real dialogue without this step.

B. Safety Gatekeeper and Access Control

SAGE security architecture in the three-layer Safety Gatekeeper is shown in **Figure ??**. Herein, we discuss access control for each candidates query, where the retrieved data chunks are represent by $d \in \mathcal{D}_{cand}$:
Role-Based Access Control (RBAC): Each of the retrieved chunks for the query carries a type label by design such as lecture note, practice solution, exam key, etc. The access to these chunks are permitted if:

$$\text{RBAC}(d, R) = \begin{cases} 1, & \text{if role\_perm}(R, \text{type}(d)) = \text{allow}, \\ 0, & \text{otherwise.} \end{cases} \tag{5}$$

For instance, exam answer keys are always denied for role $R = student$, thus removing sensitive content from possible exposure.
Attribute-Based Access Control (ABAC): Subsequently, if the retrieved chunks d corresponds to topic t, access to them depends on curriculum state and mastery i.e., for example whether the topic is open or not.

$$\text{ABAC}(d, K) = \begin{cases} 1, & \text{if status}(t) = \text{Open} \wedge \text{Ready}(t, K) = 1, \\ 0, & \text{otherwise.} \end{cases} \tag{6}$$

The instructors may lock topics or enforce pacing constraints, ensuring zone-of-proximal-development alignment.
Cognitive Access Control (CAC): We map four anchors into a conservative access decision by checking whether the retrieved document's *response mode* is compatible with the inferred intent. Towards this, a lightweight intent model infers a coarse tutoring intent label $C \in \{conceptual, problem\_solving, debugging, review\}$:

$$CAC(C, d) = \begin{cases} 1, & \text{if } mode(d) \in \mathcal{M}(C) \\ 0, & \text{otherwise} \end{cases} \tag{7}$$

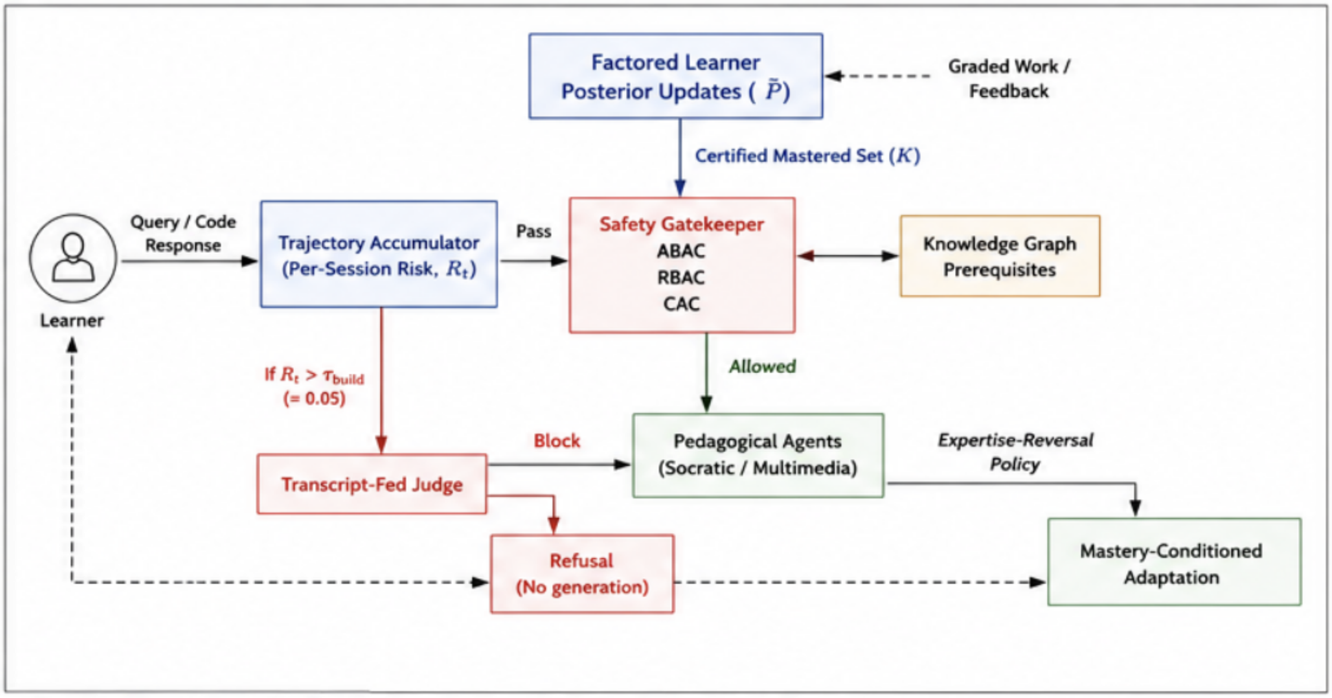

Fig. 2. End-to-end system architecture. The learner model updates posteriors from graded work and provides the certified mastered set K to the gatekeeper for ABAC decisions.

Fig. 2: End-to-end system architecture. The learner model updates per-tier posteriors from graded work and supplies the certified mastered set K to the gatekeeper for ABAC decisions; sessions exceeding $\tau(q_t)$ are adjudicated by a transcript-fed judge before the gate.

where $\mathcal{M}(C)$ encodes organizations' tutoring policy for example, *problem_solving* permits hints or partial code, while instructor-only solution keys remain disallowed for students. This formulation subsumes the binary distinction between *solution-seeking* and *scaffolding-oriented* requests by treating *problem_solving* as the solution-adjacent anchor and the remaining intents as scaffolding-oriented.

Implementation Details: The CAC layer performs low-latency intent classification using the CPU-based sentence embedding model `all-MiniLM-L6-v2` with 384-dimensional embeddings. The intent is inferred via cosine-similarity matching against coarse anchors for conceptual inquiry, problem solving, debugging, and review, with conservative heuristic overrides for security-sensitive queries. Domain-specific entity extraction is handled separately using `gliner_small-v2.1`, a zero-shot span classification model that supports custom labels without retraining. This enables robust identification of programming constructs and error types that are poorly handled by conventional NER methods. CAC decisions are intentionally conservative and composed conjunctively with RBAC and ABAC constraints. As a result, misclassification can affect the instructional responses, for example level of scaffolding or explanation depth, but does not affect content access or retrieval, which is primarily governed by RBAC and ABAC.

Context-Gated Classification: CAC narrows which documents reach the generator; the security overrides can refuse the query outright. A blunt formulation of them therefore over-refuses core curriculum vocabulary. Terms such as "infinite loop" and `fork()` denote concepts an introductory C learner must study in order to recognize and avoid, yet a phrase match alone blocks "why does my code have an infinite loop?", a debugging request, as a security risk. A false block in a tutoring system carries a direct pedagogical cost that a generic deployment does not incur: the learner is refused instruction on material the course itself requires them to master. We therefore gate these overrides on co-occurrence rather than on presence. A curriculum term is treated as adversarial only when it appears together with a harmful goal such as *crash*, *freeze*, or *never stop* in the same message, while unambiguous attack phrases such as *fork bomb* are blocked regardless of context. For the same reason, the overrides are keyed on explicit rules rather than on embedding similarity, since a fuzzy security anchor false-matches innocuous tokens and reintroduces the false blocks the gating is intended to remove. Allowing a benign single turn to proceed is what permits the session-level layer of Section III-C to adjudicate the conversation in which it sits.

As a result of the three-layered Safety Gatekeeper, a document d is admitted to the LLM context only if

$$\text{Allow}(d \mid S) = \text{RBAC}(d, R) \wedge \text{ABAC}(d, K) \wedge \text{CAC}(C, d) \quad (8)$$

A document is therefore included in the context window iff $\text{Allow}(d \mid S) = 1$. Once the context is defined, the system delegates control to different pedagogical agents that control information presentation to the learner and provide regulatory control to instructors.

C. Trajectory-Level Access Control

Each of RBAC, ABAC, and CAC in Eq. (8) evaluates a single candidate document against a single query. This posture is blind to crescendo attacks [17], in which a prohibited objective is decomposed across turns and the assembling instruction is superficially benign. Herein, we adapt the peak-and-accumulation approach of [19], which combines the worst single turn with the persistence of suspicion across turns, to the access-control boundary of an instructor-regulated tutor. The per-message gate is retained unchanged, and a session-level risk state is layered over it to determine when a conversation-level adjudication is performed.

At turn t, a scalar risk $r_t \in [0, 1]$ is computed from signals the pipeline already produces: i) the goal-alignment gap $(1 - s_{\text{goal}})$, ii) drift from the course domain, iii) accumulated off-topic strikes, and iv) proximity to attack vocabulary. No additional inference is introduced. The session retains two scalars, updated as:

$$\text{peak}_t \leftarrow \max\left(\lambda_{\text{peak}}\, \text{peak}_{t-1},\ r_t\right), \tag{9}$$

$$\text{acc}_t \leftarrow \min(1,\ \lambda_{\text{acc}}\, \text{acc}_{t-1} + r_t), \tag{10}$$

with $\lambda_{\text{peak}} = 0.9 > \lambda_{\text{acc}} = 0.8$, so that the peak cools more slowly and a single sharp excursion remains visible for several subsequent turns, and the session risk is their equal combination:

$$R_t = \tfrac{1}{2}\, \text{peak}_t + \tfrac{1}{2}\, \text{acc}_t. \tag{11}$$

The two terms capture complementary attack shapes: a single sharp excursion and a slow accumulation of individually unremarkable turns. While $R_t < \tau_{\text{judge}}$, the session proceeds under the per-message gate alone. Once $R_t \geq \tau_{\text{judge}}$, the turn is escalated to a conversation-level judge that reads the last six turns of the transcript, so that a benign final instruction is adjudicated in the context of the conversation that produced it.

Block Poisoning: A per-message block returns before the session state is updated, so a blocked malicious turn would otherwise contribute nothing to accumulated risk. The session state is therefore updated on the blocked turn before the response returns.

Trigger versus Classifier: The accumulator is a trigger and not a classifier. The per-turn risk does not separate adversarial from legitimate sessions, since a debugging session saturated with loop-termination language accumulates risk within the range attained by attacks. The design does not require separation: the accumulator determines when adjudication is worth its cost, and the transcript-fed judge performs the classification. No session is refused without conversation-level adjudication. The accumulator selects when the judge is consulted; it never decides the outcome. Every threshold in this layer is therefore a cost boundary and none is a correctness boundary, and no score, at any value, is sufficient to refuse a learner.

Request-Form Escalation: A trigger built on harm vocabulary is nonetheless blind to the attack it most needs to catch. An attack assembled slowly is, by definition, a sequence of turns containing no incriminating terms, so no threshold on a lexical score can reach it: the signal is absent rather than the threshold set too high. We therefore add a second escalation path keyed on the form of the request rather than on the vocabulary of the harm. An adversary may paraphrase a prohibited goal without limit, but a conversation that assembles an artifact must eventually ask for the artifact, and that request is invariant under paraphrase of the goal. Let $B(q) \in \{0, 1\}$ be a deterministic predicate matching construction requests, for example a request to produce a complete program, to combine previously discussed parts, or to render a description as compilable code. The escalation threshold is then request-dependent:

$$\tau(q_t) = \begin{cases} \tau_{\text{build}}, & B(q_t) = 1, \\ \tau_{\text{judge}}, & \text{otherwise}, \end{cases} \qquad \tau_{\text{build}} < \tau_{\text{judge}}, \tag{12}$$

with $\tau_{\text{build}} = 0.05$ and $\tau_{\text{judge}} = 0.45$, and escalation occurs when $R_t \geq \tau(q_t)$. A build-triggered escalation submits the transcript rather than the turn alone. This is essential: a construction request read in isolation is frequently a legitimate exercise, and only the arc that precedes it is diagnostic. The path therefore raises recall at the trigger and leaves precision with the component that has it, which is the division of labor the trigger-versus-classifier framing asserts and which the vocabulary-keyed scorer alone did not realize.

The value $\tau_{\text{build}} = 0.05$ was set after inspecting a containment failure in an earlier audit, and it is not calibrated to that failure. At 0.05 essentially any construction request in a session carrying nonzero accumulated risk escalates, so the parameter encodes a policy that construction requests are always worth adjudicating rather than an estimate of where a boundary lies. What limits its cost is that escalation cannot block, so the price of a low threshold is judge invocations rather than refusals. Section IV-D reports the resulting over-refusal rate on benign sessions and the proportion of interceptions this path accounts for on attacks it was not designed against.

The layer is inexpensive by construction. Its entire state is two scalars per session, and it introduces no model call on the common path, since B is a fixed pattern set and the judge is consulted only above threshold. This preserves the negligible security overhead of the inherited gate.

D. Multi-Agent Pedagogical Interaction

We use the term *multi-agent* to denote a set of role-specialized pedagogical agents orchestrated by the Safety Gatekeeper, rather than fully autonomous agents with independent goals or learned policies. The system comprises three role-aligned agents: i) a *Socratic Student Agent*, which generates structured hints, logic plans, and optionally flow diagrams rather than complete solutions, so that the learner performs the reasoning the task requires, ii) an *Administrative Teacher Agent*, which operates with elevated privileges, enabling curriculum manipulation such as locking and unlocking modules and access to restricted assets via secure function calls, and iii) a *Multimedia Support Agent*, which generates visual scaffolds such as state diagrams, memory models, and control-flow charts to assist conceptual reasoning and neurodiverse learners.

Because agent selection is itself governed by the session state S, the Safety Gatekeeper acts as a learning-aware controller rather than a generic content filter: which agent responds, and with what latitude, follows from the learner's role and certified mastery rather than from the surface form of the query.

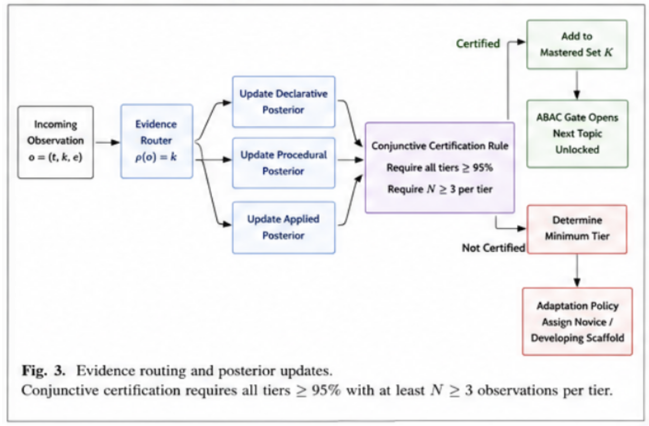

Fig. 3: Evidence routing and conjunctive certification. Each observation updates exactly one tier under ρ; certification requires every tier to reach $\theta^{(k)}_{\text{cert}}$ with at least $N_{\min}$ observations.

E. The Learner Model: Factored Knowledge Tracing

Thus far, the mastered set K has been treated as given. In SAGE as originally formulated, membership in K is conferred by course completion or instructor status, so the gate is sound but the attribute it acts on is asserted. We now specify the learner model that *earns* K from a stream of mixed interactions. Its output is the same object the gate already consumes, i.e., the set K of Eq. (1), read by $\text{Ready}(t, K)$ in Eq. (4) and by $\text{ABAC}(d, K)$ in Eq. (6), so the model changes the *provenance* of K without altering a single gate equation. We design the model to three ends: it must update online as the learner works, it must distinguish kinds of evidence rather than pooling them, and it must certify mastery under a rule an instructor-regulated gate can trust.

Mastery of a programming concept is not monolithic: recalling a definition, writing a single correct line, and debugging working logic are distinct competencies. We therefore track each topic $t \in T$ not with one estimate but with three, one per cognitive tier $k \in \{1, 2, 3\}$ corresponding to declarative, procedural, and applied understanding. Each tier maintains an unconstrained Bayesian Knowledge Tracing posterior [20] $\tilde{P}_t^{(k)} \in [0, 1]$, the probability that the learner has mastered topic t at tier k.

Evidence Routing: Each observation the learner produces is an assessed interaction $o = (t, k, e)$, where t is the topic, $e \in \{0, 1\}$ the graded outcome, and $k = \rho(o)$ the tier assigned by a deterministic routing function ρ. The routing is by instrument rather than by inference: a multiple-choice or short-answer item routes to the declarative tier, a single-line completion or trace to the procedural tier, and a submitted program or debugging task to the applied tier. The function ρ is total and single-valued, so every observation updates exactly one tier and no observation updates two. This is the property on which the guarantee of Section III-F1 rests.

Per-Tier Update: Let $P_t^{(k)}$ denote the posterior for topic t at tier k before an observation, and let the tier carry parameters $(p_G^{(k)}, p_S^{(k)}, p_{L_0}^{(k)}, p_T^{(k)})$ for guess, slip, prior, and transition. On an observation routed to tier k, the posterior is first conditioned on the outcome:

$$P_{t|e=1}^{(k)} = \frac{P_t^{(k)}\,(1 - p_S^{(k)})}{P_t^{(k)}\,(1 - p_S^{(k)}) + (1 - P_t^{(k)})\,p_G^{(k)}}, \tag{13}$$

$$P_{t|e=0}^{(k)} = \frac{P_t^{(k)}\,p_S^{(k)}}{P_t^{(k)}\,p_S^{(k)} + (1 - P_t^{(k)})\,(1 - p_G^{(k)})}, \tag{14}$$

and then advanced by the learning transition, which is applied on correct responses only:

$$\tilde{P}_t^{(k)} \leftarrow \begin{cases} P_{t|e}^{(k)} + \left(1 - P_{t|e}^{(k)}\right) p_T^{(k)}, & e = 1, \\ P_{t|e}^{(k)}, & e = 0. \end{cases} \tag{15}$$

Crediting learning after an incorrect attempt would permit wrong answers to drift the posterior upward, so the transition is withheld in that case.

The emission parameters are set per tier rather than shared, since the tiers differ in how they can be answered without competence. Table II gives the deployed values and motivates their orderings. The transition rate alone is uniform, since we have no principled basis for supposing that learning proceeds faster at one level of understanding than another.

Retention and Decay: Knowledge lapses when unpracticed. Before each update, tier k's posterior is decayed toward its prior according to the time Δt (in days) since the last interaction on the topic:

$$\tilde{P}_t^{(k)} \leftarrow \tilde{P}_t^{(k)} e^{-\lambda_k \Delta t} + p_{L_0}^{(k)} \left(1 - e^{-\lambda_k \Delta t}\right), \tag{16}$$

with per-tier half-lives of 14, 7, and 5 days for the declarative, procedural, and applied tiers, so that applied competence is treated as the most perishable. The rate λ_k is not fixed. Each successful retrieval updates an SM-2 ease factor $EF \geq 1.3$ from a quality score [29], and the rate is slowed for well-retained material as $\lambda_k \leftarrow \lambda_k / EF$, so repeatedly demonstrated mastery decays more slowly. We

TABLE II: Per-tier learner-model parameters. The orderings are structural rather than fitted: guess rates fall across the tiers, since recognition items admit more guessing than working code, while slip rates rise, since an applied task affords more ways to err despite knowledge. Priors fall likewise, since incoming learners plausibly carry partial declarative exposure but near-zero applied skill. The transition rate is uniform across tiers. Ceilings weight the composite display estimate of Eq. (17), sum to 0.95 by construction, and are excluded from certification.

Parameter	Declarative	Procedural	Applied
Guess $p_G^{(k)}$	0.20	0.10	0.05
Slip $p_S^{(k)}$	0.10	0.15	0.20
Prior $p_{L_0}^{(k)}$	0.30	0.05	0.01
Transition $p_T^{(k)}$	0.09	0.09	0.09
Decay half-life (days)	14	7	5
Ceiling $\theta_{\max}^{(k)}$	0.60	0.25	0.10

adopt SM-2 to make the decay rate responsive to retrieval history rather than uniform across learners. Combined with the hysteresis of Section III-F, this makes K non-monotone: a certified topic left unpracticed can decay below θ_{dec} and be decertified, so the gate continues to reflect current rather than historical competence.

Composite Display Estimate: For presentation to the learner, the three posteriors are summarized as a single composite,

$$\bar{P}_t = \sum_k \theta_{\max}^{(k)} \tilde{P}_t^{(k)}, \qquad \textstyle\sum_k \theta_{\max}^{(k)} = 0.95, \tag{17}$$

where the per-tier ceilings $\theta_{\max}^{(k)}$ of Table II act as weights, bounding each tier's contribution so that no single evidence type dominates. The declarative tier is weighted most heavily, which suits a progress indicator a learner consults often but is precisely the weighting the Evidence Diversity Problem warns against for a mastery judgment. The composite is therefore a presentation quantity only, and no choice of $\theta_{\max}^{(k)}$ can admit a topic to K.

F. Conjunctive Certification and the Earned Mastered Set

A topic is certified as mastered only when *every* tier is satisfied at once. Formally, $t \in K$ iff

$$\underbrace{\forall k: \ \tilde{P}_t^{(k)} \geq \theta_{\text{cert}}^{(k)}}_{\text{all tiers confident}} \ \wedge \ \underbrace{\forall k: \ N_t^{(k)} \geq N_{\min}}_{\text{all tiers evidenced}}, \tag{18}$$

with $N_{\min} = 3$ and a per-concept, per-tier certification bar $\theta_{\text{cert}}^{(k)} \geq 0.95$, floored at that value and raised where an instructor requires greater confidence for a particular concept. Theorem 1 quantifies over all admissible bars, so tightening cannot weaken the guarantee. To prevent oscillation at the boundary, certification is governed by Schmitt-trigger hysteresis: once certified, a topic is *de*certified only if some tier decays below $\theta_{\text{dec}} = 0.75 < \theta_{\text{cert}}^{(k)}$.

The certified set $K = \{t \in T : t \text{ satisfies Eq. (18)}\}$ is precisely the mastered set of the session state $S = \{K, R, C\}$ in Eq. (1). It is consumed unchanged by $\text{Ready}(t, K)$ (Eq. (4)) and $\text{ABAC}(d, K)$ (Eq. (6)); the Safety Gatekeeper is not modified. The learner model's sole effect on access control is to replace an asserted K with an earned one, so that every prerequisite and pacing decision the gate makes now rests on demonstrated, evidence-diverse competence.

1) The Evidence Diversity Guarantee

The certification rule of Eq. (18) is not merely a conservative threshold. Together with the routing of Section III-E it yields a structural property: certain evidence profiles cannot certify a topic at all, irrespective of how much evidence they contain and irrespective of the parameter values in Table II. We state this precisely.

Lemma 1 (Tier Isolation)**.** *For any observation sequence O and any tier k, the trajectory of $\tilde{P}_t^{(k)}$ and of $N_t^{(k)}$ depends only on the subsequence $O^{(k)} = \{o \in O : \rho(o) = k\}$. Observations routed to tiers $j \neq k$ leave both quantities unchanged.*

Proof. By construction of the update. An observation o with $\rho(o) = j$ applies Eqs. (13)–(15) to $\tilde{P}_t^{(j)}$ and increments $N_t^{(j)}$ alone; no term in either expression references the state of any other tier, and ρ is single-valued, so o contributes to exactly one subsequence. The decay of Eq. (16) is likewise applied per tier with per-tier rate λ_k. Hence the tier-k state is a function of $O^{(k)}$ only. □

Theorem 1 (Evidence Diversity Guarantee)**.** *Let O be any observation sequence for topic t and let $n_k = |O^{(k)}|$ be the number of observations routed to tier k. Then:*

(i) *If $n_k < N_{\min}$ for some tier k, then $t \notin K$, for every assignment of the parameters of Table II and for every $|O|$.*

(ii) *If $O^{(k)}$ contains no correct observation, then $t \notin K$, again for every assignment of the parameters of Table II and irrespective of n_k.*

Proof. For (i), Eq. (18) requires $N_t^{(k)} \geq N_{\min}$ for every k. By Lemma 1, $N_t^{(k)} = n_k$, since only observations in $O^{(k)}$ increment it. If $n_k < N_{\min}$ the second conjunct of Eq. (18) fails and certification is refused. No observation outside $O^{(k)}$ can increment $N_t^{(k)}$, so enlarging O elsewhere cannot repair the deficit, and the argument uses no property of the parameters.

For (ii), suppose $O^{(k)}$ consists entirely of incorrect observations. Under Eq. (15) no learning transition is applied, so the tier posterior evolves solely by conditioning on $e = 0$. Non-degeneracy of the emission parameters, $p_S^{(k)} + p_G^{(k)} < 1$, makes this map a strict contraction toward zero, so the sequence is monotone decreasing from $p_{L_0}^{(k)}$ and converges to 0. The decay of Eq. (16) pulls toward $p_{L_0}^{(k)} < \theta_{\text{cert}}^{(k)}$ and so cannot raise the posterior above any bound it does not already exceed. Hence $\tilde{P}_t^{(k)}$ never attains $\theta_{\text{cert}}^{(k)}$, the first conjunct of Eq. (18) fails, and $t \notin K$. □

Corollary 1 (Minimum Diverse Evidence)**.** *Certification of a topic requires at least $N_{\min}$ observations in each of the* *three tiers, and therefore at least* $3N_{\min}$ *observations in total. Under the deployed parameterization the two conjuncts of Eq.* (18) *bind simultaneously: three consecutive correct observations carry the declarative, procedural, and applied posteriors to* 0.982*,* 0.981*, and* 0.989 *respectively, so nine correct observations, three of each kind, is the exact minimum for certification.*

Remark: Both parts hold for every admissible parameterization, which permits Section III-I to defend the constants by their consequences rather than by fit. The guarantee is a property of the architecture, i.e., of the routing, the conjunction, and the withheld transition. The withheld transition is essential to part (ii): were p_T applied irrespective of outcome, an all-incorrect sequence would converge to a nonzero fixed point, 0.103, 0.108, and 0.114 for the three tiers, and part (ii) would hold only under the additional condition that these lie below $\theta^{(k)}_{\text{cert}}$.

G. Mastery-Conditioned Response Adaptation

Certification governs what the tutor may *retrieve*; the same mastery signal also governs how it should *respond*. The expertise-reversal effect [27] holds that the heavy scaffolding which helps a novice becomes redundant, and eventually obstructive, as competence grows. We operationalize this with a policy Π that maps the learner's mastery on the current topic to a response configuration, i.e., scaffolding depth, explanation style, and challenge type. Let $\ell(t) \in \{\textsc{Novice}, \textsc{Developing}, \textsc{Proficient}, \textsc{Reviewing}\}$ be the adaptation level for topic t, assigned by the same conjunctive logic that governs certification, applied to the raw per-tier posteriors:

$$\ell(t) = \begin{cases} \text{REVIEWING}, & t \in \bar{K}, \\ \text{PROFICIENT}, & \min_k \tilde{P}^{(k)}_t \geq 0.75 \ \wedge \ \min_k N^{(k)}_t \geq 2, \\ \text{DEVELOPING}, & \min_k \tilde{P}^{(k)}_t \geq 0.35 \ \vee \ \max_k N^{(k)}_t \geq 2, \\ \text{NOVICE}, & \text{otherwise}, \end{cases} \tag{19}$$

with the cases evaluated in order and $\bar{K} \supseteq K$ denoting the set of topics certified at any point in the learner's history. A topic that decays out of K therefore retains its REVIEWING treatment, so that lapsed mastery prompts revision rather than a return to introductory scaffolding, while the prerequisite gate continues to act on K itself.

The policy reads the minimum across tiers and not the composite of Eq. (17). Partitioning the composite would be compensatory by construction, and would award reduced scaffolding to precisely the learner the Evidence Diversity Problem describes: a learner at $(0.95, 0.75, 0.55)$ attains a composite of 0.81, above any threshold that would otherwise separate the levels, while their weakest tier stands at 0.55. Adaptation is therefore conjunctive for the same reason certification is, and Eq. (17) governs display alone. The asymmetry between the two clauses is deliberate: promotion to PROFICIENT requires both conditions, since withdrawing scaffolding from an unready learner is the costlier error, whereas leaving NOVICE requires either. The cutoffs gate no formal quantity and are stated as configurable policy, since the expertise-reversal literature establishes that support should fade with competence but fixes no threshold at which it should do so [27]. Table III gives the resulting response configuration at each level.

TABLE III: Mastery-conditioned response policy Π, assigned by Eq. (19). The Reviewing level is entered on certification and is not withdrawn by decay.

Level	Scaffolding & style	Challenge
Novice	Maximum; real-world analogies, concept in small steps	Guided fill-in-the-blank
Developing	No analogy; step-by-step mechanism trace, with a dedicated *Common Mistakes* section	Multi-concept synthesis
Proficient	Minimal; skips definitions, targets edge cases, memory safety, subtle bugs	Debugging / output prediction
Reviewing	None; concise refresher only	Quick retention check

One further effect ties the policy back to the gate. When a topic is certified, withholding its solution is no longer pedagogically warranted, since the learner has demonstrably mastered it, so the Reviewing level relaxes the Socratic withholding that CAC (Eq. (7)) applies to uncertified topics, replacing hint-only responses with a brief refresher and a retention check. Certification thus has two coordinated effects: it opens the prerequisite gate for *downstream* topics through K (Eqs. (4), (6)), and it lifts solution-withholding on the *mastered* topic itself.

H. Telemetry and Instructor Control

Every signal above is derived from an append-only telemetry stream that SAGE persists per session: for each concept and tier, the live posteriors, evidence counts $N^{(k)}_t$, and decay rates; a turn log recording code complexity, latency, inferred intent, and the session-risk quantities R_t, peak_t, and acc_t; and a prediction log pairing each pre-update posterior with the observed outcome for continuous Brier-score monitoring. The turn log is written on every terminal event, including blocks, so that the contribution of each enforcement layer can be attributed rather than inferred, and the append-only form supports offline replay without disturbing the live model.

Two further quantities are derived from this stream and surfaced to the instructor rather than consumed by the response path. *Cognitive load* is estimated from three telemetry signals, each normalized to $[0, 1]$, i.e., code complexity $\hat{c}$ (an AST-depth proxy), response latency $\hat{\ell}$ (capped at 20 s), and recent error rate $\hat{\epsilon}$:

$$C_{\text{load}} = 0.4\,\hat{c} + 0.3\,\hat{\ell} + 0.3\,\hat{\epsilon}. \tag{20}$$

Active misconceptions, i.e., recurrent and specific error patterns such as confusing = with ==, are tracked per concept with an exponential moving average over an indicator of confident error:

$$M_c \leftarrow \lambda_M M_c \ + (1-\lambda_M) \\ \mathbf{1}[e=0 \ \wedge \ \text{conf} \geq \theta_{\text{conf}}]\,, \quad \lambda_M = 0.7. \tag{21}$$

Confidence is self-reported on a five-point scale at the time of submission and the threshold is $\theta_{\text{conf}} = 4$, so only the upper two response categories qualify. Conditioning on confidence distinguishes a misconception from an ordinary error: a learner who is unsure and wrong is still learning, whereas one who is confident and wrong holds a belief that will not self-correct.

The same substrate carries a per-decision security record. Every gatekeeper block emits a deterministic reason naming the layer that fired, for example goal-bounded security, harmful code, semantic adjudication, or trajectory risk, appended together with the inferred intent, a query excerpt, and the prevailing session risk R_t. The instructor is thus presented with a reviewable security ledger rather than an opaque refusal, and can distinguish a learner who encountered a curriculum boundary from one who attempted to evade it. The record also supports offline red-team replay across a deployment.

This substrate extends SAGE's existing human-in-the-loop control (Sec. **??**) from curriculum authoring to learner monitoring. Where instructors could already lock topics and edit prerequisites, the learner model now surfaces, per learner and per topic, certified and decertified mastery, stalled progress, active misconceptions, and elevated cognitive load, prioritizing learners for attention accordingly. Actions taken through this interface, i.e., adjusting pacing, unlocking a prerequisite, or flagging a misconception for review, propagate immediately to the gate, closing the loop between what the model observes and what the instructor regulates.

I. Implementation and Parameter Selection

The system is implemented as containerized microservices, with FastAPI orchestrating ChromaDB [30] and Neo4j [31] as synchronized services and all policy enforcement centralized in the Safety Gatekeeper. Intent detection and access checks run on CPU-based local models rather than remote LLMs, reducing security overhead to 135 ms and Time-to-First-Token from 13.1 s to 2.2 s. Baseline systems use GPT [32] and Gemini [33] via public APIs without fine-tuning; SAGE uses Gemini 2.5 Flash as its generator but routes every query through the Safety Gatekeeper and controlled retrieval pipeline before generation.

The constants introduced above fall into three kinds, defended by different standards. For most of them there is no reason the value is 0.10 rather than 0.12, and we do not manufacture one. What can be defended is that the value lies in a principled range and that the conclusions are stable across it, which is why the sensitivity analyses of Section IV-F are part of the argument rather than an appendix to it.

Policy thresholds, i.e., the certification bar $\theta^{(k)}_{\text{cert}}$, the evidence count $N_{\min}$, the decertification bound θ_{dec}, and the escalation thresholds τ_{judge} and τ_{build}, are not estimated quantities. They encode the confidence an institution requires before content is unlocked or an adjudication is performed, are instructor-configurable, and are justified by their consequences in the manner of a confidence-level convention. The provenance of τ_{build} in particular is given in Section III-C.

Model parameters, i.e., the per-tier guess, slip, and prior rates and the decay constants, are engineering defaults whose orderings are motivated where each is introduced in Section III-E. Their specific values affect only secondary quantities, since the Evidence Diversity Guarantee holds for all admissible parameter values and Section IV-F reports the stability of certification across the plausible region. *Display constants* shape only what is shown to the learner and are excluded from certification by construction. We do not defend infrastructure conventions such as retrieval breadth or decoding temperature, which bear on none of the results reported here.

IV. Performance Evaluation

We conduct a multi-dimensional evaluation of our SAGE ITS system to test its effectiveness for secure and instructor-regulated learning *without* degrading instructional value. We analyze: (i) *Response Quality*: Does the system actually guide the student through the learning process, or does it simply give away the answer? We check if the LLM adheres to the curriculum and offers hints instead of solutions, (ii) *Safety and Security*: Can the system be tricked? We test if students can bypass the rules to access hidden exam keys or content they are not supposed to see yet, and (iii) *System Performance*: Is the system optimized enough for real-time user interaction? In all experiments, SAGE is compared against two state-of-the-art commercial LLM baselines (GPT and Gemini). These two LLMs are selected as they are two of the top models in the LLM leaderboard [34].

A. Experimental Setup

We evaluate along four modalities. First, a controlled simulation harness generates synthetic learners with known ground-truth mastery. Each learner is assigned a latent per-tier mastery vector, and responses are sampled with tier-appropriate guess and slip rates. We distinguish balanced learners, whose mastery is uniform across tiers, from lopsided learners, who are strong on declarative recall but weak on applied competence, the profile the Evidence Diversity Problem concerns. The harness imports the deployed update function and evidence configuration directly rather than reimplementing them, so the behavior measured is that of the running system.

Second, a fixed educational benchmark is replayed through the deployed system and scored against the conference configuration, and the adversarial suites of Section IV-D are collected against a live deployment. Neither runs in the simulation harness: both exercise the full request path, and each response is recorded together with the system state that produced it.

Third, we conduct supervised, task-based usability sessions with individual participants. Each session comprises i) a pre-session questionnaire establishing prior experience and self-reported confidence, ii) a fixed task battery exercising the mechanisms of Section III under think-aloud, and iii) a post-session questionnaire pairing standardized usability instruments with the confidence items repeated from the pre-session form. Sessions are conducted one participant at a time on a dedicated machine, so that every participant meets the same tasks under the same conditions.

Fourth, the system is deployed on commodity cloud infrastructure and used unsupervised over 6 weeks by a cohort of 10 participants engaged in genuine coursework on their own machines. Three sources are collected: the append-only telemetry stream of Section III-H, which records every interaction; a rolling participant-maintained issue log, which time-stamps difficulties as they are encountered; and a closing questionnaire recording overall experience. Each logged issue resolves by timestamp to its session in the telemetry stream, so a reported difficulty can be examined against the system state that produced it. We treat the two sources asymmetrically: rates and distributions are computed from telemetry, whereas the issue log establishes which behaviors participants noticed and when, and is a lower bound on occurrence rather than a census. The two participant modalities answer different questions: the supervised arm establishes how the system behaves on tasks we control, and the unsupervised arm establishes which behaviors surface when we do not.

Judged quantities are scored by an LLM judge [35], for which we adopt two controls. Every verdict records the judge model that produced it, and every verdict is content-hashed against the response it scored, so that a verdict cannot be reapplied to a response it never saw when a suite is re-collected. We report the judge model alongside every judged figure and cross-check the principal judged results against a second judge from a different model family. Unless stated otherwise, simulation results report 2000 learners per cell at a fixed seed with Wilson 95% intervals, and paired comparisons use McNemar's exact test. Results from the two participant modalities are reported descriptively with participant counts.

B. Metrics

The metrics that we use to evaluate responses from SAGE and baseline LLM-based AI-Tutor systems are as follows: *a) Code Density:* The fraction of response content that appears as executable code (e.g., code blocks or code-like lines). Lower values indicate the tutor avoids dumping full solutions and instead emphasizes guidance. *b) Security Compliance:* The percentage of Security-category benchmark prompts for which the system refuses or safely redirects disallowed requests. *c) Curriculum Compliance:* The percentage of queries for which the response respects course constraints such as locked topics, prerequisites, and instructor-defined pacing. *d) Pedagogy Score (1–5):* A rubric-based rating where a score of 5 reflects stepwise scaffolding and a score of 1 reflects direct solution dumping. *e) False Certification Rate:* The proportion of simulated learners certified for a topic on which their true per-tier competence does not meet the certification bar. Reported as a raw count, since the rates of interest are near zero and a percentage obscures the denominator. *f) Deferral Rate:* The proportion of simulated learners genuinely competent on all tiers who are not certified. Reported separately from (e), since the two error classes carry different costs and are not interchangeable. *g) Π-Fidelity:* The agreement between the adaptation level assigned by the policy and the level recovered by a blind judge shown only the generated response, reported with Cohen's κ against a chance baseline of $1/|\ell|$. *h) Attack Success Rate:* The proportion of adversarial multi-turn sessions in which harmful content was delivered at any turn, judged on delivery rather than on the presence of a refusal message. Scoring every turn rather than only the terminal payload charges the system for content extracted along the way, so the figure lower-bounds containment. *i) Benign Over-Refusal Rate:* The proportion of legitimate multi-turn sessions blocked at any turn. Reported alongside (h) rather than combined with it, since the deployed base rate is not the balanced rate of the audit suite.

C. Pedagogical Alignment on C-EduBench

We first establish that the learner model is integrated without degrading the instructional behavior of the inherited system. Table IV compares raw and prompt-engineered baselines against the conference system and the present extension on C-EduBench.

Two access-control results follow directly from the learner model. Curriculum compliance rises from 80.0% to 100.0%, the expected consequence of replacing an asserted mastered set with an earned one: prerequisite decisions that previously rested on course enrollment now rest on per-tier evidence, and the residual violations of the conference system were concentrated in exactly those cases. Security compliance rises from 9 of 10 to 10 of 10 refusals on the Security subset; at $n = 10$ we report this as an absence of regression rather than as a gain ($p = 1.0$).

The instructional metrics require one stratification. The replay runs on an account with no certified prerequisites, so the gate returns a prerequisite roadmap rather than an answer for 23 of the 50 items, and all 10 Boundary items are gated by construction. A roadmap contains no code and teaches nothing directly, so both response metrics are read on the 27 items where the tutor taught.

Under that stratification the pedagogy score rises from 2.37 to 4.59, a paired gain of 2.22 points ($p = 1 \times 10^{-5}$, $n = 27$), against a pooled gain of 1.68 points across all 50. Code density moves the other way: the pooled reduction of 6.4 points falls to 1.3 points on the taught subset, which is not distinguishable from noise at this sample size and is carried almost entirely by the gated stratum. We therefore report pedagogy as the instructional result and not density, since gating is already counted as curriculum compliance and crediting it twice would attribute to the response policy an effect produced by the gate.

Against prompt-engineered baselines the comparison is unchanged in character from the conference version. Prompt-based tutors remain competitive on pedagogy, with GPT (Tutor) at 4.0, while enforcing policy at a rate a course cannot rely on: 30.0% security compliance against 100.0%, and no curriculum enforcement to measure. This gap is not one better prompting closes, since the baselines fail on retrieval-boundary decisions their prompt has no authority over.

D. Multi-Turn Enforcement

Per-message enforcement cannot see an attack assembled across a conversation, so we audit the session-level layer separately. The suite comprises 40 multi-turn adversarial sessions spanning eight strategies: slow-burn, paraphrase-evasion, academic-framing, keyword-sparse, crescendo, role-play, social-engineering, and decomposition. Fifteen were used during development of the layer, including the containment failure that motivated the request-form path of Eq. (12). The remaining 25 were authored after that path was fixed and without reference to the patterns it matches. Nine benign multi-turn sessions, one of them adversarially worded but legitimate, serve as controls. The entire suite was collected and judged three times to establish run-to-run variance.

Table V reports the outcome. The system deflects 170 of 171 adversarial turns and contains 39 of 40 sessions, delivering a complete artifact in fewer than one session in twenty and containing both restricted-asset attempts. Containment is not deterministic: three attacks, one each from the slow-burn, paraphrase-evasion, and keyword-sparse classes, delivered in exactly one run of three and were contained in the other two, and no attack failed twice. All three are held-out; every development attack was contained in every run. We therefore report the mean with its observed range rather than the best run. The single benign false block is unchanged across every configuration we test, and the adversarially worded control escalates to the judge and is cleared.

Attribution: Because per-turn session risk is recorded on every terminal event, including blocks, the contribution of each layer is measured rather than inferred. Of 61 interceptions, the escalated transcript judge accounts for 31, with

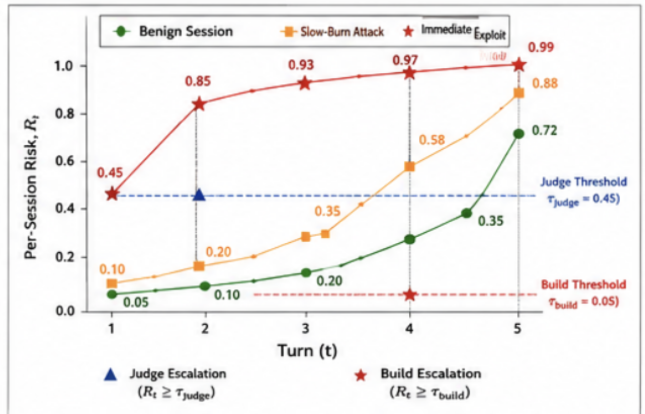

Fig. 5. Risk trajectories (R_t) over turns for benign and attack sessions. Escalations occur at judge threshold (τ_{judge}) and build threshold (τ_{build}).

Fig. 4: Per-session risk over turns. Benign sessions reach the range attacks occupy, which is why the pooled score does not separate the two populations; escalation nonetheless occurs on every attack, and the benign session that crosses τ_{judge} is cleared by the judge.

per-message layers accounting for 13 goal-bounded, 14 off-topic, 2 harmful-code, and 1 attention-hijacking block. Every interception attributable to the session layer occurs at a recorded risk above the threshold that applied to it, while every per-message interception occurs at a recorded risk of exactly zero, since those layers fire on the first turn, before any trajectory has accumulated. This is also why the pooled risk score does not separate attack sessions from benign ones, at an AUC of 0.491 ($p = 0.89$) per turn and 0.596 ($p = 0.47$) on the session peak:

the pooled statistic averages a conversation-level mechanism over turns at which that mechanism is definitionally inactive. The two-stage design predicts this, and the attribution record confirms it.

The request-form path generalizes: Restricted to the 25 attacks authored after it was fixed, 9 of 16 judge interceptions fired at a session risk below τ_{judge}, between 0.066 and 0.39, and are therefore escalations only the request-form path can produce; the path was the first block on 8 of those 25. Five held-out slow-burn attacks were intercepted this way at risks between 0.066 and 0.104. A mechanism fixed against a single observed failure thus caught five attacks it had never seen.

Ablation: To isolate the layer we raise both escalation thresholds above the attainable range of R_t, leaving the score computed and logged but never acted upon, and leaving every per-message layer untouched. The manipulation check holds exactly: goal-bounded and harmful-code blocks stand at 13 and 2 in every arm, and every turn whose block state changes is one the session judge had blocked. Containment falls from 39 of 40 to 34 of 40, and payload-turn interception from 32 to 19. Six sessions differ, all in the predicted direction; on the four cleanly attributable pairs McNemar gives $p = 0.125$, the smallest value attainable at this sample size. The effect is large and consistently signed, and the

TABLE IV: Comparative performance across raw, prompt-engineered, conference, and extended systems on C-EduBench. Lower is better for code density; higher is better for security compliance, curriculum compliance, and pedagogy score. Judged rows use `gpt-4o`. The code-density column is decomposed in the text: most of the reduction is attributable to prerequisite gating rather than to the response policy.

Metric	GPT (Raw)	GPT (Tutor)	Gemini (Raw)	Gemini (Tutor)	SAGE	IRL-EXT
Code Density (↓)	10.0%	**6.9%**	23.0%	9.4%	16.6%	10.2%
Security Compliance (↑)	20.0%	30.0%	20.0%	30.0%	90.0%	**100.0%**
Curriculum Compliance (↑)	NA	NA	NA	NA	80.0%	**100.0%**
Pedagogy Score (1–5) (↑)	3.1	4.0	3.4	3.8	2.22	**3.90**

TABLE V: Multi-turn adversarial audit over 40 attack sessions and 9 benign controls. Containment is judged on delivery of harmful content at any turn, not on the presence of a refusal message. Payload-turn gating is lower than session containment because four attacks evaded the gate and nonetheless delivered nothing, as discussed in the text. Figures are the mean of three collect-and-judge cycles, with the observed range where runs differ. ↑ / ↓ indicate that higher / lower is better.

Outcome	Result
Sessions with no harmful delivery (↑)	39/40 (97.5%) [95, 100]
Adversarial turns deflected (↑)	170/171 (99.4%)
Complete artifact delivered (↓)	0.7/40 (1.7%) [0, 2]
Payload turn gated (↑)	32/40 (80.0%)
Restricted-asset containment (↑)	2/2 (100%)
Benign sessions never blocked (↑)	8/9 (88.9%)

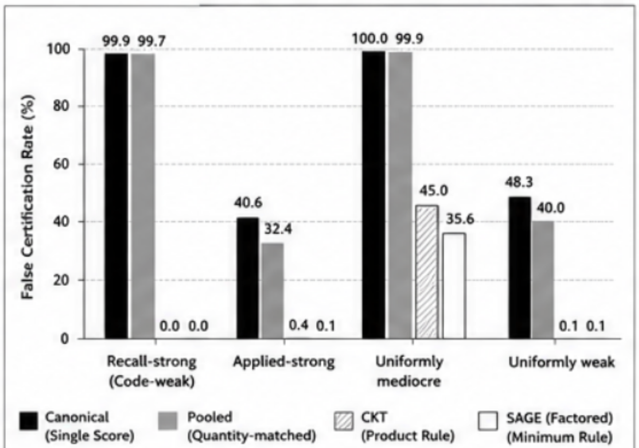

Fig. 4. False certification rate (%) across learner archetypes (Table VI). Lower is better.

Fig. 5: False certification rate across learner archetypes, from Table **??**. Lower is better. Both single-score rules certify nearly every recall-strong learner; the two multi-dimensional rules separate only on the uniformly mediocre archetype.

sample cannot establish it.

The ablation also shows the two layers composing as designed. Four slow-burn attacks that the session layer gates before retrieval remain contained without it, because generation-time refusal catches them a turn or two later. The session layer's distinct contribution is therefore to intercept at the gate what the generator would otherwise refuse after retrieval, and to close the residual deliveries that generation-time refusal misses, approximately one per twenty attacks.

Containment that is not enforcement: Four paraphrase-evasion attacks were never gate-blocked and nonetheless delivered nothing. Their payload turns were stripped of context to evade both the harm lexicon and the request-form patterns, and were consequently so unspecific that the tutor returned a generic concept roadmap or misread the request. The evasion is self-defeating in a narrow sense: abandoning the specificity that triggers the gate also abandons the specificity needed to elicit a working artifact. We report these as contained but not as enforced, since an adversary who found phrasing both trigger-evading and context-preserving is not excluded by this result.

E. Over-Crediting Resistance

Decomposing quantity, diversity, and compensation: Our rule differs from standard knowledge tracing in two respects that these archetypes can separate: it factors evidence by tier, and it conjoins the tiers under a minimum rather than a product. We evaluate four rules on identical response streams. The first is the canonical formulation, in which mastery is declared when the posterior crosses θ_{cert} with no evidence requirement [20]. The second is a strengthened baseline that pools evidence into a single posterior and additionally requires the nine observations Eq. (18) demands in total; it is included solely to isolate the effect of evidence quantity. The third is conjunctive knowledge tracing [24], operationalized over the same per-tier posteriors as a product $\prod_k \tilde{P}^{(k)} \geq \theta_{\text{prod}}$. Sharing the posteriors ensures that any difference from our rule is attributable to the certification rule alone, and $\theta_{\text{prod}} = 0.95^3 = 0.857$ is the value at which the two rules coincide when every tier sits exactly at θ_{cert}, so the comparison is not rigged by threshold placement. The fourth is the factored rule of Eq. (18).

Table **??** reports the result. Across the four under-prepared archetypes, 2000 learners each, the canonical rule falsely certifies 5776 of 8000 and the factored rule 716, a reduction of 87.6%. The factored rule's false-certification rate is never higher than any alternative on any archetype, and is strictly lower than every alternative on two. Three effects account for this.

The first is dimensionality, and it dominates. On the recall-strong, code-weak archetype the Evidence Diversity Problem names, both single-scalar rules certify essentially every learner, at 1998 and 1994 of 2000, while both multi-dimensional rules certify none. An existing conjunctive criterion achieves this too, so the elimination is attributable to dimensionality rather than to our rule. What the strengthened pooled baseline establishes is that quantity is not the fix: demanding the same total volume of evidence without separating its kinds reduces false certification by four learners in two thousand. The failure is one of evidence diversity, not of evidence quantity.

The second effect is compensation, and it is where the two multi-dimensional rules separate. On the uniformly mediocre archetype, whose true competence sits near 0.65 on every tier, the product rule certifies 899 of 2000 and the factored rule 712, a reduction of 187 learners, or 21% of the product rule's false certifications. The same ordering holds on the applied-strong archetype, where the factored rule certifies a quarter as many, at 0.1% against 0.4%. The mechanism is visible in the rule: a product lets two strong posteriors carry a weak third past a fixed threshold, whereas a minimum requires each tier to clear its bar independently and admits no such offset. A product rule could be retuned to match the observed rate at one operating point. What retuning cannot alter is that the offset remains available at every product threshold, while the minimum forbids it at all of them. That is the content of Theorem 1, and it is the property an access-control decision requires, since a gate that opens on compensated evidence admits a learner to downstream content on the strength of competence they were never shown to possess.

Both multi-dimensional rules nonetheless certify a substantial share of this archetype, which Theorem 1 does not address: a learner competent at nothing in particular exhibits no lopsidedness for a conjunction to detect, and the certifications arise from posterior drift under a nonzero transition rate over 60 observations. The remedy is calibration of the certification bar and the transition rate rather than the form of the conjunction, and Section IV-F reports the sensitivity of both.

The third effect is the cost. The factored rule certifies genuine masters about four points less often than the product rule, at 73.8% against 77.9%, and its errors on the master archetype are deferrals. The two error classes are not interchangeable: a deferral withholds certification from a competent learner and is recoverable with further evidence, whereas a false certification opens a curriculum gate that should have stayed closed. We accept a higher rate of the recoverable error for a lower rate of the unrecoverable one.

F. Decision Quality: Deferral versus Misclassification

On the recall-strong, code-weak archetype the two rules fail in different directions. The factored rule produces no false certification and errs only by deferring true masters; the pooled rule certifies 1994 of 2000 under-prepared learners and defers none. The errors fall into different classes rather than differing in rate, and the classes are not interchangeable.

Deferral is a property of the curriculum, not of the rule: The right panel of Fig. **??** reports the proportion of true masters certified as a function of the share of applied items in the assessment mix. At a 5% share the factored rule certifies 34.1% of true masters, at 10% it certifies 73.8%, and at 20% and above it certifies 97.4% or higher. Deferral therefore arises when the curriculum supplies too little applied evidence for the applied tier to accumulate, and not from conservatism in the rule. The enrichment that removes the cost does not weaken the guarantee: at a 20% applied share the pooled baseline still certifies approximately 99% of recall-strong, code-weak learners, so the additional evidence benefits the factored rule alone.

The rule therefore imposes a requirement on assessment design rather than on the learner. The configuration used here supplies a 10% applied share at a 26.2% deferral rate on true masters. We recommend an applied share of at least 20%, at which the guarantee is obtained at a deferral cost below 3%.

Sensitivity to the emission parameters: We perturb the two parameter families the result could plausibly depend upon, the guess and slip rates and the transition rate, over four settings each, and re-run every archetype at 2000 learners per cell. Across the eight resulting parameterizations the factored rule falsely certifies recall-strong, code-weak learners *twice* in 16,000 trials, against 15,813 for the canonical rule and 15,776 for the quantity-matched pooled rule. No configuration produced a violation of the certification inequality. The separation does not depend on the values in Table II; it follows from requiring evidence in every tier, which no parameter assignment can relax. The perturbations vary the parameters of the emission model while retaining its form, so the result is established as insensitive to parameter choice within the assumed model; Section V treats misspecification of the model itself as the principal threat to the simulation evidence.

The pooled baseline uses parameters averaged over the item mix, at $p_G = 0.155$, $p_S = 0.125$, $p_{L_0} = 0.196$, and $p_T = 0.09$.

G. Mastery-Conditioned Adaptation

A learner model is consequential only if it changes what the learner receives. We therefore ask whether the same query yields materially different responses across proficiency states, and whether that difference is recoverable by an observer who does not know the learner. Six synthetic learners were written directly into the knowledge store, differing only in their per-tier posteriors. Three fall at the DEVELOPING level under Eq. (19) and one at each of NOVICE, PROFICIENT, and REVIEWING. The DEVELOPING level is over-represented by design, since it spans the band where the minimum and the composite of Eq. (17) disagree most sharply, e.g., the learner at $(0.95, 0.75, 0.55)$ of Section III-G. Each was issued the same 20 queries on a single concept in a fresh session, giving 120 responses through the live pipeline.

The adaptation policy assigned the correct treatment on all 120 turns, verified against an independent transcription of the

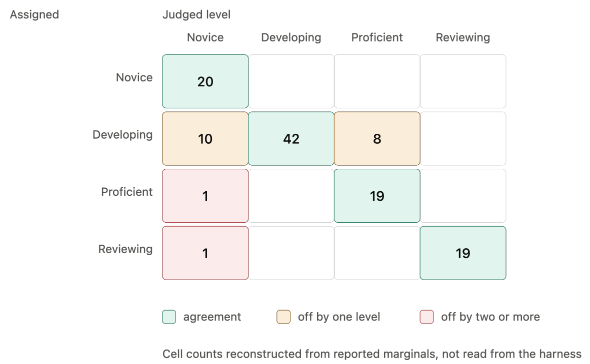

Fig. 6: Assigned adaptation level against blind-judge recovery ($n = 120$). Shaded cells are exact matches. Errors concentrate on adjacent levels; the developing row carries three learners, one of which accounts for eight of its misreads.

deployed rule. Table VI illustrates the difference for a single query. Responses differ not merely in length but in rhetorical strategy: the novice treatment leads with an analogy and defines every term, the developing treatment withholds analogy and traces the mechanism, the proficient treatment omits definition entirely and opens on failure modes, and the reviewing treatment introduces no new material. Section structure differs correspondingly, each level emitting at least one section no other level produces.

Fidelity: A blind judge shown a single response, with no information about the learner, recovered the assigned treatment in 83.3% of cases (100/120, 95% CI [75.7, 88.9]; $\kappa = 0.767$) against a 25% chance baseline, with no response declined. Errors are ordinal rather than arbitrary, at 18 of 20 falling between adjacent levels and a mean ordinal distance of 1.15.

Deterministic features separate the levels in the same direction, with code density rising from 33.9% at novice to 61.4% at proficient, analogy cues falling from 1.55 to 0.05 per response, and every adjacent pair separating on at least one feature. These counts are instructed by the prompt, so they measure compliance; the blind judge measures fidelity.

Where the policy is least sharp: Recall is 100% at novice and 95% at proficient and reviewing, but 70% at developing, and the confusions concentrate on one learner profile. A learner weak on recall but strong on applied work is correctly assigned the developing treatment by the conjunctive rule, yet the generated response is judged proficient in 8 of 20 cases, since a code-strong learner receives a code-heavy answer that presents as advanced. The disagreement is between the policy and the generation layer, and it is a limit of the present adaptation prompt rather than of the assignment rule.

Weak-tier identification: Where evidence supports it, the system names the learner's weakest tier, returning the applied tier for recall-strong learners and the recall tier for applied-strong learners. The name is withheld when the spread between tiers is small or the weakest tier is unevidenced. Both conditions are load-bearing: an unconditional minimum returns the applied tier for every learner, since it carries the lowest prior and is therefore weakest by construction rather than by evidence.

H. Beta Testing result

I. Usability Testing result

V. Discussion

Factoring a learner model by evidence type and certifying conjunctively removes the substitution a pooled scalar admits. Dimensionality does most of that work, and an existing conjunctive criterion achieves it too; what the factored rule adds is a conjunction that is non-compensatory at every threshold rather than at a tuned one. The distinction matters for instructor-regulated learning rather than for measurement, since the certified set is the attribute on which every retrieval-time access decision turns: an inflated record is not merely a measurement error but an escalation of privilege, and the content it unlocks here includes practice solutions and restricted assessment material. The session-level layer supports a narrower claim by design, since the accumulator is a trigger that routes a conversation to a transcript-fed judge and the judge is the classifier. Two points generalize beyond this system. Wherever evidence of differing cost and guessability is aggregated into one quantity and thresholded, the cheap evidence can purchase the credential, and compensation survives any monotone aggregation that is not a minimum. And a treatment a prompt instructs cannot be verified by counting the behavior the prompt instructed, so recovering the assigned level from response text alone is the manipulation check that precedes any question of effect.

Four directions follow. A controlled comparison over a full term would establish whether mastery-conditioned adaptation improves learning outcomes, which no arm here measures, and would run long enough to observe the decertification the decay model implies. Generating responses under a materially different process and certifying under the deployed rule would test the guarantee against misspecification of the emission model rather than only its parameters, which is the highest-value extension of the simulation evidence. A larger adversarial suite would narrow the confidence intervals, give the ablation the power this sample cannot, and extend coverage beyond the strategies we anticipated. Validating judge verdicts against human raters would close the principal unaddressed threat to every judged figure we report.

VI. Conclusion

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TABLE VI: Mastery-conditioned adaptation. The same query, "Can you explain what a pointer is in C?", issued by four learners differing only in their per-tier BKT state. Excerpts are verbatim system output.

Level	Words	Rhetorical move	Excerpt
Novice	316	Analogy-led; defines every term	"…a variable is like a labeled box where you store a piece of data…Think of it like a scavenger hunt: if a normal variable is the treasure, a pointer is a piece of paper with the exact GPS coordinates of where that treasure is hidden."
Developing	363	No analogy; traces the mechanism	"A pointer is a memory address stored in a variable, but the power of pointers lies in the interaction between the address-of operator (`&`) and the dereference operator (`*`)…you are instructing the CPU to look at the address stored in `ptr` and perform a read or write directly at that memory location."
Proficient	258	No definition; opens on failure modes	"Since you are comfortable with basic pointer mechanics, you should be aware of the risks regarding pointer arithmetic and memory safety. Accessing memory via an uninitialized pointer…results in undefined behavior, which often manifests as intermittent segmentation faults or silent data corruption."
Reviewing	165	Teaches nothing new; reminds	"Remember that a pointer is simply a variable that stores the memory address of another variable…A common 'gotcha' that often fades is the difference between the pointer's address and the value it points to."

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